Task E3 — OTA Strategy Selection

Full replacement, canary and shadow mode for a 280 KB model | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component E, Module 5

At a glance

Item	Result
Recommendation	**Shadow mode on all 85 trucks**: the new model runs beside the old one for 7 days without acting, then is promoted by a small config message
Cost per update cycle	Full replacement ₹2.38 (23.8 MB); canary ₹2.43 (₹0.61 if rolled back); shadow ₹3.20 with on-device summaries, ₹53.80 if every window is streamed
Why cost does not decide it	The whole spread is under ₹52 per cycle. One vaccine-spoilage loss was ₹28 lakh, about 52,000 times the dearest option.
Why shadow	It exposes no truck to an unvalidated model (full: 85, canary: 10), compares the models on every window of every truck, and needs no connectivity for a rollback because the new model never acted

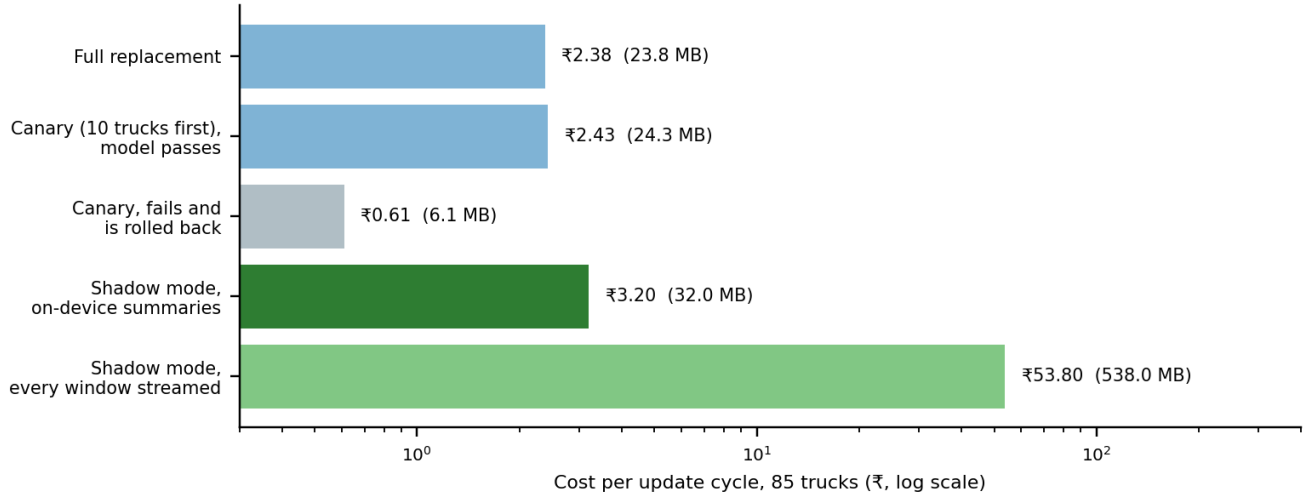
1 Inputs

From the brief	Value
Model file	280 KB (INT8 TFLite)
Update cadence	Every 6 weeks (8.67 cycles a year)
Fleet	85 trucks in the pilot, 265 at full scale
Tariff	₹0.10 per MB on an M2M SIM

My assumptions (not in the brief)	Value
Units	1 KB = 1,000 B and 1 MB = 1,000,000 B, as in Task A1
Canary and shadow observation period	7 days
Inference windows	One every 10 s, 8,640 a day (Task C2)
Hourly summary report	300 B per truck
Disagreement record	150 B each; 0.5% of windows disagree
Streaming every window	100 B per window
Promote message	1 KB per truck
Package manifest overhead	Ignored; about 10 KB per truck (Task D2) would add under 4%

2 Bandwidth cost per update cycle

Strategy	Model MB	Telemetry MB	Total MB	₹ per cycle	₹ per year	₹ per cycle, 265 trucks
Full replacement	23.80	0.00	23.80	2.38	20.63	7.42
Canary, model passes	23.80	0.50	24.30	2.43	21.06	7.47
Canary, fails and is rolled back	5.60	0.50	6.10	0.61	5.29	0.61
Shadow, on-device summaries	23.80	8.22	32.02	3.20	27.75	9.98
Shadow, every window streamed	23.80	514.16	537.97	53.80	466.24	167.72



For scale, one vaccine-spoilage loss was ₹28 lakh (₹28,00,000), about 52,000 times the dearest bar.

Workings

Strategy	Calculation
Full replacement	85 × 0.28 MB = 23.80 MB; × ₹0.10 = ₹2.38
Canary, model passes	(10 + 75) × 0.28 MB = 23.80 MB, plus 10 trucks × 7 days × 7.2 KB of hourly reports = 0.50 MB: 24.30 MB, ₹2.43
Canary, fails	10 trucks out and 10 back = 5.60 MB, plus 0.50 MB of reports: 6.10 MB, ₹0.61. The other 75 never receive it.
Shadow, summaries	23.80 MB, plus 85 × 7 days × 13.68 KB (7.2 KB summaries + 6.48 KB of disagreements) = 8.14 MB, plus 85 × 1 KB = 0.09 MB: 32.02 MB, ₹3.20
Shadow, streaming	23.80 MB, plus 85 × 7 days × 864 KB = 514.08 MB, plus 0.09 MB: 537.97 MB, ₹53.80

3 The cold-chain and rural facts that decide it

Fact	Consequence
All 85 pilot trucks carry cold-chain pharmaceuticals, and one refrigeration-failure loss (₹28 lakh of vaccine) happened in 18 months	There are no “safe” trucks to test on, and a missed fault is very costly
That is about one real fault per 78 weeks across the fleet. A 7-day canary on 10 trucks has a 0.15% chance of meeting one; shadow on 85 trucks, 1.3%	Real faults are too rare to confirm a model catches them. Whatever the strategy, validation needs a way to test faults without waiting for one, and without letting an untested model decide
Task D1: the model passed the 88% gate with 98.25%, yet missed 3.9% of anomalous windows when a fault started mid-run	A model that passes offline checks can still be unsafe in the field, so it should not act until it is compared on real data
Coverage gaps of 35–90 minutes at seven places on the Nashik–Aurangabad route	Reports and rollback commands reach a truck late, so a bad model keeps running unmonitored for up to 90 minutes per gap
280 KB takes 45 s at 50 kbps and under 4 minutes at 10 kbps	The transfer fits any coverage window. Connectivity affects monitoring and rollback far more than download

4 Recommendation: shadow mode

Send the 280 KB model to all 85 trucks and keep the old model resident. For 7 days both run on every window; only the old model raises alarms. Each truck compares the two on the truck, sends an hourly summary and a short record for each disagreement, and uploads them when coverage returns. The ops centre promotes the new model, by a small config message, only if these gates pass:

- **Safety disagreements:** zero windows where the new model says Normal and the old one says Warning or Critical, or every one is reviewed and explained.
- **Overall disagreement:** at most 1% of windows (my suggested threshold).
- **Fault replay:** recorded and simulated faults, including faults that start mid-run, are injected into the new model on the truck or in test and must all be detected.
- **Output stability:** the PSI monitor from Task E1 stays below 0.10 for the new model on every truck.
- **Coverage:** summaries received from at least 95% of trucks, since gaps delay reports.

The price is ₹0.82 more per cycle than full replacement (₹7.13 a year), plus a second inference per window: about 0.03% of the CPU on a Raspberry Pi 5 (2.8 ms per 10 s window, Task B2).

5 Why not the other two

Full replacement

- **Every truck is exposed at once.** A model with a blind spot like the one in Task D1 would blind all 85 pharmaceutical loads together, and a single missed refrigeration failure is a ₹28 lakh event.
- **Rollback depends on connectivity.** A truck inside a 90-minute gap keeps running the bad model, and trucks update at different times anyway, so the fleet is never uniform.
- **It saves almost nothing.** Shadow costs ₹0.82 more per cycle.

Canary (10 trucks first)

- **It lets 10 real pharmaceutical loads depend on an unvalidated model.** If the model misses a fault on one of them, no alarm is raised and the load can spoil.
- **A clean canary week proves little.** With about a 0.15% chance of any real fault on those trucks, “nothing went wrong” does not show that faults are caught.
- **The canary is unrepresentative and slow to report.** Easy-coverage routes hide the rural conditions where trouble is likeliest, and reports from the canary arrive late after each gap.
- **It is nearly as expensive as shadow** (₹2.43 against ₹3.20) while protecting far less.

Canary's real strength is that it tests the new model's alerts in real operations. If that matters, run a canary after shadow mode, not instead of it.

6 Decision matrix

Criterion	Full replacement	Canary	Shadow (recommended)
Trucks exposed to an unvalidated model's decisions	85 (100%)	10 (12%)	0
Real-data comparison with the old model	None	10 trucks, if at all	All 85 trucks, every window
Judged on	Field outcomes, which are rare	Field outcomes, which are rare	Agreement with the old model plus fault replay
Rollback needs connectivity	Yes, urgently	Yes, for 10 trucks	No: the new model never acted
Bandwidth per cycle	₹2.38	₹2.43	₹3.20
Time to benefit	Days	About 1 week, then the rest	About 1 week, then a config switch
On-truck complexity	Lowest	Low	Higher: two models and a comparison

7 Weaknesses of shadow mode and how they are covered

Weakness	Cover
The benefit arrives about a week later (7 of 42 days, 17% of the cycle)	Acceptable for a routine 6-week update. An urgent safety fix is a separate decision and needs sign-off.
Agreement with the old model is not correctness	Add the fault-replay gate, and review every safety disagreement by hand.
Telemetry can balloon if every window is streamed (₹53.80 per cycle)	Aggregate on the truck: ₹3.20 per cycle.
Coverage gaps delay the summaries	Store and forward as in Task A2; decide on the reports that have arrived and require 95% coverage.
More code on the truck	Keep the old model as a standby slot, so promoting or reverting is a config switch with no download.

8 Items to confirm

- **Assumptions.** The 7-day soak, 0.5% disagreement, 300 B hourly summary and 150 B disagreement record are mine. `ota_strategy_cost.py` recomputes everything if you change them.
- **Kilobyte definition.** I used 280 KB = 280,000 B. With 1 KiB = 1,024 B the model is 286,720 B and every figure rises by 2.4%.
- **Fleet scale.** The right-hand column shows the 265-truck cost; the ranking does not change.
- **Your stack.** Shadow mode needs the truck to hold two models and compare them. Tell me if your runtime cannot, and I will revisit canary with extra safeguards.